AGENTVERSE 2026 · ROBOTICS

Rig Robot

One quadruped, two jobs on an offshore rig: rust repair and gas-leak response. It knows when to call a human.



THE PROBLEM

Two chronic jobs that put people in the hazard zone

Corrosion

Pipework rusts constantly in salt spray. People fix it by hand from rope access, 30 m up, next to live hydrocarbons.

Small gas leaks

Bolted flanges work loose. A person with a gas detector walks up to the leak to find it.

Dangerous · costly · never finished. User: asset-integrity and maintenance crews.

The robot must never **paint over rust** or **put a wrench on a cracked flange.**

WHAT WE BUILT

Go2 + 0–0.45 m lift mast + UR3 arm, one brain

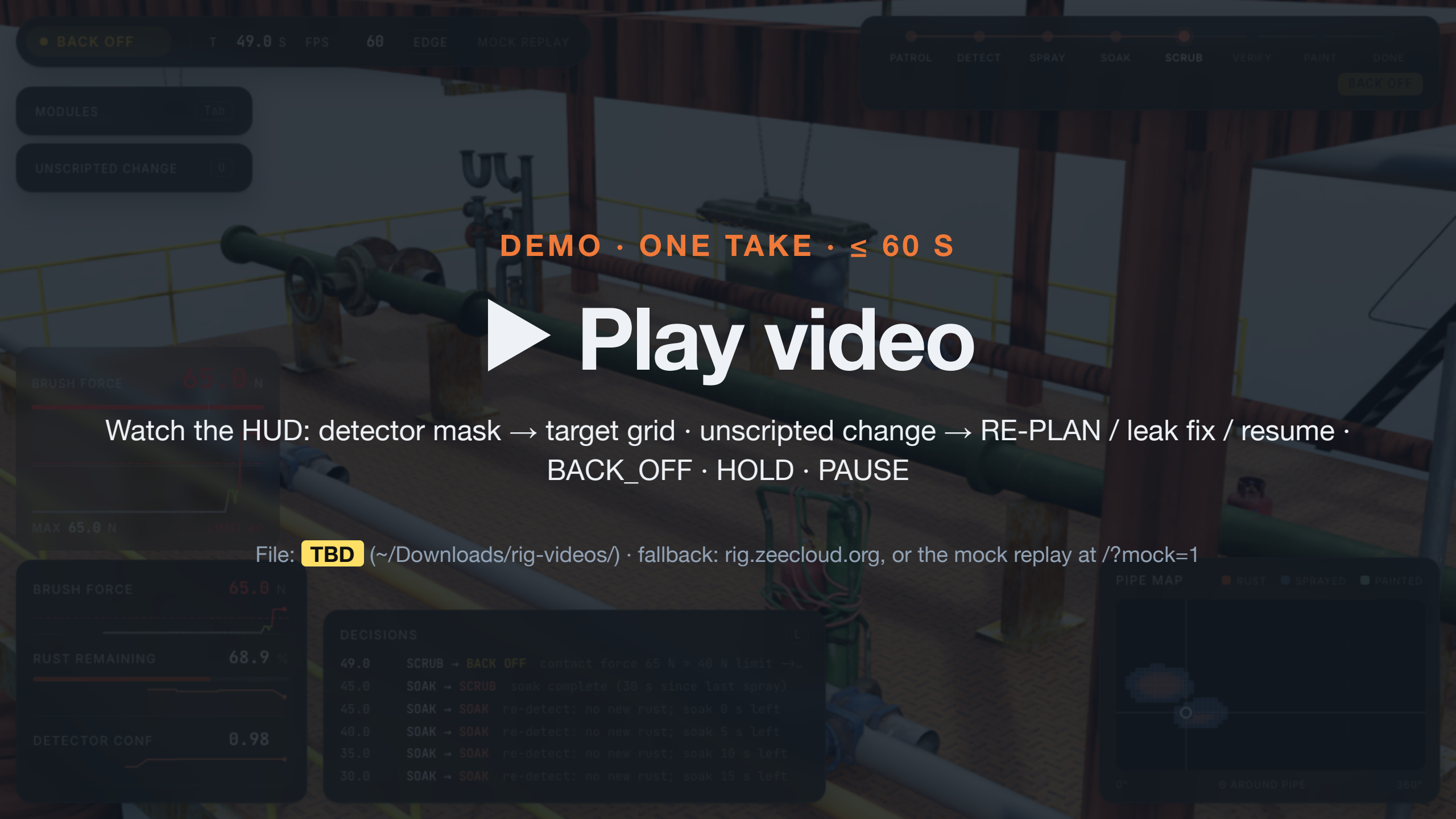
On a full offshore rig in Isaac Sim 6.2, rendered on an NVIDIA GB10 and streamed live with a HUD.

RUST DETECT → SPRAY → SOAK 30 s → SCRUB · PPO → VERIFY · detector + Gemini → PAINT

LEAK GAS_CHECK → LEAK_APPROACH → LEAK_INSPECT → RETORQUE → LEAK_VERIFY → resume

ANY GAS_RETREAT · BACK_OFF · SECOND_VIEW · HOLD · PAUSE_PERSON · ESCALATE_TO_HUMAN

The gas check runs at every patrol stop and before every step: **gas pre-empts everything**. One state machine, one camera, one arm, one decision log.



DEMO · ONE TAKE · ≤ 60 S

▶ Play video

Watch the HUD: detector mask → target grid · unscripted change → RE-PLAN / leak fix / resume · BACK_OFF · HOLD · PAUSE

File: **TBD** (~/Downloads/rig-videos/) · fallback: rig.zeecloud.org, or the mock replay at /?mock=1

PERCEPTION-DRIVEN ACTION

Pixels become the work plan

RUST · MOBILENETV4-S + LORA, 1.1 MS ON CPU

Frame → 96 px patches → Heatmap → Ray–cylinder → (θ, z) grid on pipe → Spray · scrub · done check

0.977
F1, 881 real photos

13/13
held-out Isaac renders

LEAK · NO ML: TORQUE STRIPES + CRACK CHECK

Frame → Project known bolts → Read each red stripe → Thin dark shapes, mm length → View quality → Loose · crack · unsure

40/40
cracks found

0/40
false alarms

≈ 1°
bolt-angle error

SCRUB POLICY · PPO IN ISAAC LAB, 2,048 ENVS

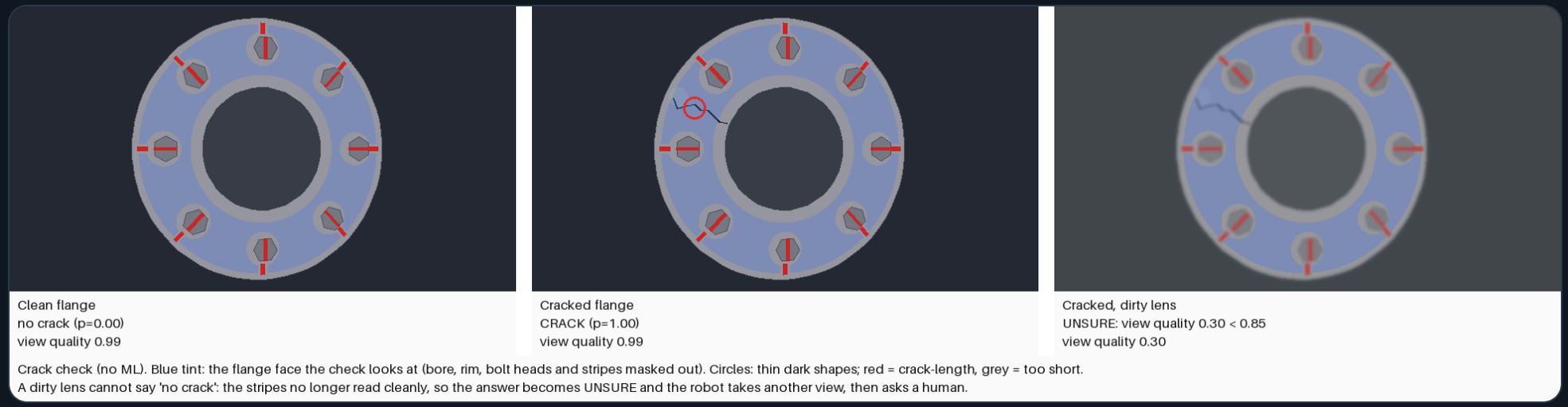
v1 cleaned everything by pressing too hard

RIG GEOMETRY, NEW RUST PATCH EACH EPISODE	V1	V2 (FORCE-LIMITED)
Success (< 5 % rust left)	100 %	97.6 %
Episodes over the 40 N limit	**76.8 %**	**1.0 %**
Mean / p95 episode-max force	**49.9 / 70.1 N**	**29.9 / 35.7 N**
Mean time to clean	5.2 s	6.8 s

Wear model: removal ~ force × speed, 4× faster after a 30 s soak. v2 adds a quadratic over-force penalty and a 38 N setpoint cap. The v2 demo render peaks at 31.9 N.

GAS-LEAK JOB

Loose bolt, or crack? The camera decides



Clean flange
no crack (p=0.00)
view quality 0.99

Cracked flange
CRACK (p=1.00)
view quality 0.99

Cracked, dirty lens
UNSURE: view quality 0.30 < 0.85
view quality 0.30

Crack check (no ML). Blue tint: the flange face the check looks at (bore, rim, bolt heads and stripes masked out). Circles: thin dark shapes; red = crack-length, grey = too short.
A dirty lens cannot say 'no crack': the stripes no longer read cleanly, so the answer becomes UNSURE and the robot takes another view, then asks a human.

Loose bolts 1 and 5

Re-torque, stripe aligned, gas clears → DONE

Cracked flange

Back off 3 m, critical alert → ESCALATE

Dirty lens

“No crack” not trusted → 2 extra views → ESCALATE

Leak mid-scrub

Pause, fix bolt 2, resume the scrub → DONE

CRITICAL FAILURE MODES

The most severe hazard always wins

Paint over rust	< 5 % rust, ≥ 80 % visible · Gemini veto · 3 cycles → ESCALATE
Wrench on a crack	crack possible → no bolt touched · back off 3 m
Gas ≥ 0.20 LEL · sensor fault	**GAS_RETREAT** (fail-safe)
Gas during rust work	pause → fix leak → resume
Force > 40 N	**BACK_OFF**: retract 5 cm, × 0.7
Person < 3 m · lens blocked	**PAUSE_PERSON · HOLD**



The HUD in mock replay: BACK_OFF at 65 N > 40 N, with the decision log and pipe map.

Check order before every step: gas sensor working → gas below limit → gas clear → person clear.

EDGE AUTONOMY · WHAT'S NEXT

The fast loop never leaves the robot

On the robot · CPU

- Rust detector: 1.1 ms
- Scrub policy: 0.012 ms*
- Leak check: ~90 ms / frame*
- GB10 TensorRT: **TBD**

Cloud · optional

- Gemini rust second opinion
- Asynchronous · latency **TBD**
- Offline → unchanged behaviour

Training only

- Modal A10G · L40S
- GB10: PPO + RTX render + stream

Coming **TBD**

Unit 2 “gas” robot: methane %LEL, H_2S, OGI-style camera, acoustic sensor, cast-and-surge source localisation · 3D plume with orifice leak rate · fleet planner with gas pre-emption · storm adaptation · Go2 walking

*measured on a laptop CPU, batch 1.



Find it. Fix it. Prove it.
Ask when unsure.

Limits: simulation only · robot posed kinematically · brush force is a spring model, bolt torque not simulated · gas thresholds illustrative · leak checks not yet re-tested on Isaac frames

rig.zeecloud.org · docs/technical_summary.pdf